Gender Classification Pipeline — Transparencia.be FOI Dataset

5-Tier Cascade: Tier 0 Greeting → Larousse → Statbel → WGND 2.0 → Qwen 2.5 LLM (Tiers 0–3 deterministic, Tier 4 near-deterministic)

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This pipeline classifies requester and respondent gender using a five-tier cascade. Tier 0 (response greeting) takes precedence for requesters because it reflects how the administration perceived the requester's gender — the signal that shapes their responsiveness, which is what this study examines. Tiers 1–3 use deterministic dictionary lookups as a proxy when no greeting is available. Tier 4 (Qwen 2.5 LLM) is a probabilistic fallback for unresolved cases.

A -SHARED INPUT

Dataset Input

4,265 FOI requests

Requests filed 2016–2025

Export Folder Structure

```
transparencia-export-full/
  ├── request_id_slug/
      ├── 2022-03-15-143000/
      │     └── outgoing_0001.txt  ← citizen's FOI request
      └── 2022-04-02-091500/
            └── incoming_0001.txt  ← official's response
```

4,265 folders found **(100%)**

Reference Databases

TIER 1	TIER 2	TIER 3
Larousse des prénoms (Ferey, 2019)	Statbel Belgian Registry (2017)	WGND 2.0 (Lax-Martinez et al., 2021)
9,641 names	**8,146** additional names	**3,482,709** names

Combined lookup: **3,486,454** unique first names

B -PARALLEL PROCESSING

First outgoing_*.txt

First incoming_*.txt

Requester Lane

R1 -FILE SELECTION

Select First Outgoing Text

Earliest-timestamped outgoing file per request

4,265 requests

R2 -BOILERPLATE REMOVAL

Strip Platform Footer[7]

Remove Transparencia.be platform boilerplate text

R3 -SIGNATURE ZONE EXTRACTION

Closing Formula Detection[1]

Scan last **500** characters for closing formula

96.9% have closing formula → extract from formula to end

3.1% fallback → last 200 characters

R4 -NAME EXTRACTION

Multi-Method Name Extraction

Respondent Lane

S1 -FILE SELECTION

Select First Incoming Text

Earliest-timestamped incoming file per request

4,265 requests checked

1,754 (41.1%) -NO REPLY received

⊘ dead end

2,511 have incoming text

2,511 with text

S2 -BOILERPLATE REMOVAL

Strip Footer & Quoted Request[7]

Remove platform footer + quoted original request text

S3 -SIGNATURE ZONE EXTRACTION

Closing Formula Detection[1]

59.8% have closing formula

- Heuristic: capitalized words on dedicated lines → **3,955**
- spaCy NER: PER entities after closing formula → **100**
- Single-name fallback: lone known first name → **60**

No name found → **150** (flow to Tier 4)

Total: **4,115** name tokens extracted **(96.5%)**

4,115 tokens

R5 -FIRST NAME VALIDATION

Dictionary Lookup

Check each token against **3,486,454** known names (≥3 chars)

Checked sequentially: Larousse first, then Statbel if not found, then WGND if still not found

→ dictionary cascade

R6 -TIER CASCADE

TIER 1 -LAROUSSE
3,529 M/F + 208 BOTH

↓ *not found*

TIER 2 -STATBEL
51 M/F + 1 BOTH

↓ *not found*

TIER 3 -WGND 2.0
97 M/F

↓ *229 names not in any dictionary*

Tiers 1–3: 3,677 M/F + 209 BOTH = 3,886 classified (91.1%)

HOW THE 4,115 EXTRACTED NAMES WERE RESOLVED:

```
3,886 matched a known first name → classified by Tiers 1–3 ✓
  229 not found in any dictionary ┐
                                  ├→ 308 sent to Tier 4
  150 no name extracted (from R4) ┘

Tier 0 cross-check: 1,305 response greetings detected (M=940, F=365)
    71 resolved cases with no name, 73 resolved unisex names, 14 overrode dictic
    (Response text is cleaned of platform boilerplate before greeting dete
```

308 remaining

TIER 4 -QWEN 2.5 (OLLAMA)[2]
308 cases (158 unmatched names + 150 no name)
Temperature=0, seed=42, anti-hallucination prompt
Results: 9 M/F | 299 UNKNOWN | 0 no result

M/F = LLM found a clear personal first name + surname in the signature zone
UNKNOWN = LLM instructed to return this when no personal name is visible (initials, organisations, pseudonyms)
No result = LLM response could not be parsed into the expected format
Full prompts shown in annotation [2]

Hallucination filter[5] (applied to 9 M/F results only): verify name exists in input text → discard if absent. UNKNOWN/no-result cases unaffected.

REQUESTER WATERFALL -ALL 4,265 CASES

```
├── 1,305 Tier 0 (Greeting)     │ perceived gender (precedence)
├── 3,529 Tier 1 (Larousse)     ┐
├──    51 Tier 2 (Statbel)      │ 3,677 M/F (name-based)
├──    97 Tier 3 (WGND)         ┘
├──   136 BOTH (209 original, 73 resolved by Tier 0)
├──     9 Tier 4 Qwen (M/F)          TO REVIEW
├──   299 Tier 4 Qwen (UNKNOWN)      = 304 unresolved (post-review)
──────────────────────
   Combined: 3,825 M/F + 136 BOTH + 304 UNKNOWN = 4,265 ✓
```

R7 -FINAL CLASSIFICATION (SEX_FINAL)

3,825	136	304
M/F -89.7%	BOTH[4] -3.2%	UNKNOWN -7.1%

9 Tier 4 cases + 14 Tier 0 disagreements flagged TO REVIEW

40.2% fallback → last 200 characters

S4 -NAME EXTRACTION

Multi-Method Name Extraction

- Heuristic: capitalized words → **1,065**
- spaCy NER: PER entities → **188**
- Single-name fallback → **21**

No name in incoming → **1,237** (flow to Tier 4)

Total: **1,274** name tokens extracted

S5 -FIRST NAME VALIDATION

Dictionary Lookup

Check each token against **3,486,454** known names (≥3 chars)

Checked sequentially: Larousse first, then Statbel if not found, then WGND if still not found

→ dictionary cascade

S6 -TIER CASCADE

TIER 1 -LAROUSSE
993 M/F + 24 BOTH

↓ *not found*

TIER 2 -STATBEL
40 M/F

↓ *not found*

TIER 3 -WGND 2.0
217 M/F

↓ *0 unmatched names (every extracted name was found in a dictionary)*

Tiers 1–3: 1,250 M/F + 24 BOTH = 1,274 classified

HOW THE 2,511 REPLIES WERE RESOLVED:

```
1,274 had a name in the text → name looked up in dictionaries → all 1,274 matc
1,237 had reply text but no personal name visible (e.g. department signatures,
```

1,237 remaining

TIER 4 -QWEN 2.5 (OLLAMA)[3]
1,237 cases with reply text but no name extracted
1,754 had no reply at all (skipped -nothing to analyze)
Results: 70 M/F | 1,164 UNKNOWN | 3 no result

M/F = LLM found a clear personal first name + surname in the signature zone
UNKNOWN = LLM instructed to return this when no personal name is visible (department names, automated replies)
No result = LLM response could not be parsed into the expected format
Full prompts shown in annotation [3]

Hallucination filter[5] (applied to 70 M/F results only): verify name exists in input text → discard if absent. UNKNOWN/no-result cases unaffected.

RESPONDENT WATERFALL -ALL 4,265 CASES

```
├── 1,754 no reply received        (no text exists)
├──   993 Tier 1 (Larousse)     ┐
├──    40 Tier 2 (Statbel)      │ 1,250 M/F
├──   217 Tier 3 (WGND)         ┘
├──    24 Tier 1 BOTH (Larousse)      = 24 BOTH
├──    70 Tier 4 Qwen (M/F)            TO REVIEW
├── 1,164 Tier 4 Qwen (UNKNOWN)   ┐
└──     3 Tier 4 Qwen (no result) ┘ = 1,167 no name in text
──────────────────────
   4,265 requests for which we checked the response ✓
```

S7 -FINAL CLASSIFICATION (SEX_FINAL)

1,290	24	2,951
M/F -30.2%	BOTH[4] -0.6%	UNKNOWN -69.2%

Among 2,511 with reply: 49.8% classified by Tiers 1–3 (30.2% full pipeline incl. Tier 4)

70 Tier 4 cases flagged TO REVIEW

Model Upgrade: Llama 3.2 → Qwen 2.5

The pipeline evolved through two iterations. An initial run with Llama 3.2 (3B) produced a high error rate, identified during manual review by a co-author. The pipeline was re-run with Qwen 2.5 (7.6B) — a 2.5x larger model and top performer in a published benchmark for few-shot clinical IE with locally deployed models — and Tier 0 (greeting detection) was added.

Llama 3.2 (3B) — initial

Tier 4 M/F cases: **164** (136 req + 28 resp)
Error rate: **74%** (122/164 incorrect)
Model guessed M/F too aggressively

Qwen 2.5 (7.6B) — final

Tier 4 M/F cases: **79** (9 req + 70 resp)
Error rate: **51.9%** (41/79 → corrected to 100%)
More conservative, fewer false M/F

164 → 79	+85	+144
M/F cases (52% reduction)	Correctly returned UNKNOWN	Resolved by Tier 0 (new)

After exhaustive manual review and 41 corrections, all Tier 4 classifications are human-verified (100% post-review accuracy). The final pipeline produces **1,198 known-sex pairs** (28.1% of 4,265 requests) available for gender-based analysis.

Tier 4 Filter Activity (Combined)

Stage	Count
LLM returned M/F with detected name	**88**
False positive filter (function word as name)	0 caught
Hallucination filter (name not in text)	**14** caught
Passed both filters	**74**
LLM returned M/F without name (bypassed filters)	**5**
Total M/F in sex_final (Tier 4)	**79 (9 req + 70 resp)**

Manual Review[6]

79 Tier 4 M/F + 14 Tier 0 disagreements = 93 cases

9 requester + 70 respondent (Tier 4) + 14 Tier 0 disagreements — all undergo exhaustive manual verification

Tier 4 review outcome: 41 of 79 corrected (51.9% pre-review accuracy → 100% post-review; all 41 fixed in the final dataset)

Corrections breakdown:
- 34 cases → reclassified as UNKNOWN
- 7 cases → gender corrections

Summary: 41 corrections applied during manual review (pre-review Tier 4 accuracy was 48.1%).

Tier 0 disagreement review: 14 cases, 0 adjustments needed

- **3 group requests:** pipeline extracted the first co-signer but the administration addressed a different one
- **11 dictionary overrides:** name classified differently from how the administration perceived the person (e.g., "Yan" = F in Larousse but admin writes "Monsieur Roosens")

In all 14 cases, Tier 0 (the greeting) correctly reflected the administration's perception.
(9 former disagreements were eliminated by improved multi-line neutral greeting detection, e.g. "Madame,\nMonsieur,")

UNKNOWN validation: Random sample of 200 from 1,462 Tier 4 UNKNOWN cases. Result: **156/200 (78%)** correctly UNKNOWN (requester: 97%, respondent: 59%). The lower respondent rate reflects conservative LLM behaviour on complex multilingual inputs and preprocessing that sometimes stripped names before the LLM could see them — resulting in reduced coverage rather than incorrect classifications.

Tier 0-3 sample validation: Random sample of 200 from 3,911 cases resolved by deterministic tiers. Requester: **196/200 = 98.0%**. Respondent: **81/119 = 68.1%**. Lower respondent accuracy stems from long bilingual disclaimers, dictionary false positives on common Dutch/French words, and separator patterns confusing name extraction — standard limitations of automated NER on unstructured multilingual institutional correspondence. The pipeline prioritizes precision over recall: conservative parameters reduce false positives at the cost of missing some identifiable names.

Annotation Notes

▼ 1 **Closing formula**

French administrative letters conventionally end with a politeness formula (e.g., *"Cordialement"*, *"Veuillez agréer mes salutations distinguées"*) followed by the signer's name. The pipeline searches the last 500 characters for such formulas and extracts everything from the formula to the end of the text -this is the "signature zone" where the author's name appears.

▶ View all 16 closing formulas used by the pipeline ↓

▼ 2 **Why 308 requester cases for Ollama**

308 cases remain after Tier 0 cross-checking resolves 71 of the original 379 unmatched cases. These 308 include 158 cases where a name token was extracted but not found in any of the three dictionaries (possibly a surname, rare name, or false positive), plus 150 cases where no name was extracted at all. Ollama receives only the signature zone text (not the entire document) to minimize the risk of picking up names from the body of the letter that belong to other people. Since the FOI data could not be sent to cloud-based LLMs for privacy reasons (Transparencia.be requested that data not be processed by external services), a locally hosted model (Qwen 2.5 via Ollama) was used. Qwen 2.5:7b was selected after an initial run with Llama 3.2 (3B parameters) revealed a high error rate upon manual review. The model runs entirely on the researcher's machine with no data leaving the local environment.

▶ View exact requester prompt sent to Ollama ↓

▼ 3 Why respondent Ollama success is higher (5.7% vs 2.9%)

The 1,237 respondent cases sent to Ollama are those where the heuristic and spaCy found NO name at all in the reply text. These are predominantly institutional signatures (*"Le Secrétariat communal"*, *"Service Urbanisme"*), automated acknowledgments, or email disclaimers -texts that genuinely contain no personal first name. With Qwen 2.5, the relationship inverts compared to the earlier Llama 3.2 run: the respondent M/F rate (5.7%) is now higher than the requester rate (2.9%), because the improved model is better at finding personal names in institutional reply texts that are longer and more varied than citizen request signatures.

▶ View exact respondent prompt sent to Ollama ↓

▼ 4 BOTH / UNKNOWN (UNISEX)

Names like Claude, Dominique, or Camille are used for both sexes in French. Larousse marks these as *"mixte"*. For Statbel, a conservative 90% threshold was applied when building the reference database: a name is classified M or F only when ≥90% of bearers in Belgian civil registry data are of that sex, ensuring high confidence. Names below this threshold are classified as BOTH (unisex) rather than M/F -they are never counted among the gendered results.

Limitation: In theory, a small number of UNKNOWN cases from the pipeline could be names that a less conservative threshold might have resolved. However, this concerns at most a handful of cases: Tier 2 (Statbel) classifies only **52 requester** and **40 respondent** cases in total -a marginal share of the overall sample, where the vast majority are already identified by Tier 1 (Larousse) and Tier 3 (WGND). The practical impact of this threshold on the final UNKNOWN count is therefore negligible.

Future integration: The 135 requester and 24 respondent BOTH cases could potentially be partially resolved through probabilistic gender assignment -a standard technique in gender studies (Blevins & Mullen, 2015; Lax-Martinez et al., 2021). Rather than discarding unisex names, each case would be weighted by the actual male/female frequency ratio in population data (e.g., if Statbel shows 62% of "Dominique" bearers in Belgium are female, the case would count as 0.62 F + 0.38 M in the statistical analysis). This was not implemented in the current pipeline, which uses a conservative binary approach, but could be explored in future iterations to recover some of the BOTH cases without sacrificing confidence.

▼ 5 Hallucination filter

This filter applies only to cases where the LLM returned a name with a gender classification (M/F or flagged for REVIEW). After the LLM returns a name, the pipeline checks that the name actually appears in the signature zone text it was given. If the name is absent (the LLM invented it), the classification is discarded and reclassified as UNKNOWN. Cases where the LLM returned UNKNOWN or produced no result are not affected -there was no name claim to verify.

▼ 6 Why only 93 cases need review

Tiers 1–3 are deterministic dictionary lookups — same input always gives same output, no review needed. Only Tier 4 (LLM) is probabilistic, so ALL 79 of its M/F outputs (9 requester + 70 respondent) are flagged for human verification. Of these, **41 were corrected (51.9% pre-review error rate)**: 34 reclassified as UNKNOWN, 7 gender corrections. Additionally, 14 Tier 0 disagreements (where the response greeting diverges from dictionary classification) were manually reviewed — breakdown: 3 group requests (first co-signer extracted but admin addressed a different one), 11 dictionary overrides. In all 14, the greeting correctly reflected perceived gender; no adjustments needed. (9 former disagreements were eliminated by improved multi-line neutral greeting detection.) Total: 93 cases reviewed. UNKNOWN cases validated separately: 200-sample from 1,462 Tier 4 UNKNOWN cases (156/200 = 78% correctly UNKNOWN; requester 97%, respondent 59%). Tier 0–3 deterministic tiers validated: 200-sample (requester 196/200 = 98.0%, respondent 81/119 = 68.1%).

▼ 7 How platform boilerplate is identified

The Transparencia.be platform appends boilerplate to every message. The pipeline uses a two-stage cleaning system:

Outgoing texts (citizen requests): 4 prefix markers (dashes, Merci d'utiliser, Gelieve dit, Please use this email).
Incoming texts (official responses): A first-stage cut using 9 markers (quoted-message headers, email forwarding headers, and 3 shared platform markers), followed by a second-stage pass using 8 additional platform footer strings plus quoted-message patterns and "References / Visible links" sections.

Everything from the first marker onward is stripped before signature zone extraction. For respondent texts, the pipeline also strips quoted original requests (detected via *"-----Message d'origine-----"*, *"Le [date] a écrit:"*, and similar email quoting conventions) to prevent the requester's signature from being mistaken for the respondent's.

▶ View second-stage platform footer markers (8) ↓

D -FINAL PIPELINE OUTPUT

Classification Summary

Final gender classifications after all five tiers and manual review.

Final Classification Counts

Metric	Requester	Respondent
Male (M)	**2,428**	**712**
Female (F)	**1,397**	**578**
M/F total	**3,825** (89.7%)	**1,290** (30.2%)
UNKNOWN (unisex)	136	24
Unresolved	304	2,951

Known-sex pairs (both requester and respondent classified M/F): **1,198** (28.1% of 4,265 requests). These are the cases available for gender-interaction analysis.